

Adelowo Adebangi Oluwatimileyin

Applied Mathematician · Scientific Machine Learning Researcher

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PROFILE

M.Sc. Mathematical Engineer working at the intersection of numerical PDEs, computational fluid dynamics, inverse problems, and scientific machine learning. Experience developing and verifying spectral and finite-element solvers, reduced-order models, PDE-constrained inverse and optimisation methods, and Bayesian/data-assimilation workflows, with numerical verification through manufactured and exact solutions, convergence studies, and independent benchmarks. Complemented by production experience building machine-learning models, signal-processing pipelines, and data systems in Python, PyTorch, NumPy/SciPy, SQL, and FastAPI.

RESEARCH INTERESTS

Numerical PDEs and scientific computing; computational fluid dynamics; inverse problems and PDE-constrained optimisation; reduced-order modelling and operator learning; scientific machine learning; dynamical systems and data assimilation; uncertainty quantification.

PROFESSIONAL EXPERIENCE

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|---------------------|---|
| Dec 2024 – Present | Data Scientist <i>Intuos Srl</i> , L'Aquila, Italy <ul style="list-style-type: none">Developed machine learning classifiers for real-time aircraft flight-phase identification.Designed FFT-based signal-processing pipelines for aircraft engine RPM estimation, frequency analysis, and anomaly detection.Built automated telemetry-processing pipelines and operational analytics workflows for flight-performance monitoring.Delivered REST APIs exposing machine learning models and engineering analytics to operational systems. |
| Apr 2023 – Nov 2024 | Machine Learning Engineer <i>N-and Italia Srl</i> , Italy <ul style="list-style-type: none">Developed predictive machine learning models for vending-machine delivery-failure forecasting using large-scale transaction and telemetry data.Performed exploratory and statistical analysis to identify patterns affecting system performance and operational reliability.Built data-processing and KPI-monitoring workflows to support data-driven operational decisions. |
| Jan 2023 – Mar 2023 | Operational Data Analyst <i>eResult Srl</i> , Cesena, Italy <ul style="list-style-type: none">Optimised SQL queries and automated recurring reporting workflows for business-intelligence systems.Developed reproducible data-processing workflows and transformed operational data into actionable performance metrics. |
| Apr 2022 – Oct 2022 | Digital Engineering Intern <i>Bridgestone EMIA</i> , Rome, Italy <ul style="list-style-type: none">Developed computational-geometry algorithms for engineering simulation and finite-element meshing workflows.Improved data-structure operations for tire cross-section geometry processing and implemented closest-point algorithms for cubic Bézier curves. |
| May 2022 – Aug 2022 | Machine Learning Intern <i>Fellowship.AI</i> , Remote <ul style="list-style-type: none">Implemented an InsetGAN architecture for whole-body human-image generation and developed the associated deep-learning inference workflow.Deployed model inference through FastAPI for integration with a Swift iOS application. |

SELECTED PROJECTS

Applied Mathematics & Scientific Computing

2D Incompressible Navier–Stokes: Pseudo-Spectral Solver and POD–Galerkin ROM

[GitHub repository](#)

- Developed and verified a Fourier pseudo-spectral solver for 2D incompressible Navier–Stokes with integrating-factor RK4 and 2/3-rule dealiasing, confirming fourth-order temporal and spectral spatial convergence against the Taylor–Green exact solution and an independent manufactured solution.
- Derived a tensorial POD–Galerkin reduced-order model with DEIM hyper-reduction, verified to machine precision against the projected full-order operators; achieved $38.5\times$ online speed-up at $<0.1\%$ state error at rank 16, and quantified a snapshot-POD failure to generalise across unseen flow realisations ($\approx 99\%$ error).

Finite-Element Solver for 2D Incompressible Flow: Schäfer–Turek Cylinder Benchmarks

[GitHub repository](#)

- Implemented a Taylor–Hood P_2/P_1 FEniCSx/PETSc solver for incompressible viscous flow, verified by manufactured solutions recovering the theoretical $O(h^3)$ velocity- L^2 and $O(h^2)$ velocity- H^1 /pressure- L^2 convergence rates.
- Reproduced the steady and unsteady DFG Schäfer–Turek cylinder benchmarks (drag, lift, pressure difference, and Strouhal number via FFT), retaining two small, explicitly documented near-misses on peak drag/lift rather than claiming full agreement.

Deterministic and Bayesian Inversion of Spatially Varying Darcy Permeability

[GitHub repository](#)

- Developed adjoint-based deterministic inversion for a spatially varying log-permeability field (UFL automatic differentiation, H^1/L^2 Tikhonov regularisation with discrepancy-principle selection, L-BFGS-B), verified by a Taylor-remainder test, and diagnosed a regularisation-model mismatch on structured (inclusion-like) truth fields.
- Extended to Bayesian inversion with a KL-parameterised Gaussian prior sampled by preconditioned Crank–Nicolson MCMC, independently validated against an analytical linear-Gaussian posterior (0.59% error); reported MAP rather than the posterior mean as the primary estimate given diagnosed low effective sample size.

Phase-Field Modelling of Boiling Heat Transfer: Toward 3D DNS

[GitHub repository](#)

- Built a conservative Allen–Cahn/Navier–Stokes/energy phase-field solver for vapour-liquid phase change (Roccon 2025), exploiting a constant-coefficient FFT pressure-Poisson solve that extends unchanged to 3D; the 2D bubble-growth benchmark agrees with the analytical growth rate, with the error decreasing systematically under grid refinement.
- Diagnosed the failure of an initial 1D Stefan-problem implementation and developed a dedicated non-periodic 1D Stefan solver that converges at first order (interface-position error 1.142% to 0.079% for $\Delta x = 2.0\text{--}0.125$ mm at $t = 250$ s, matched densities) with verified mass and energy balances; numerically verified the 3D solver (Poisson/projection convergence, phase-mass identity, 2D/3D consistency; this exposed and corrected a missing 3D viscous term). The general multiphase heat-flux pathway remains under validation and 3D physical validation is not yet done.

2D Elasticity FEM and SIMP Topology Optimisation

[GitHub repository](#)

- Verified a P_1 plane-stress elasticity solver by the method of manufactured solutions and a SIMP compliance sensitivity by a Taylor-remainder test before trusting it for optimisation; reduced canonical cantilever compliance from 617.86 to 101.39 under a volume constraint with Bourdin density filtering.
- Characterised the method’s limits directly: the mesh-resolution study is mesh-independent-looking but not quantitatively mesh-converged ($\approx 14\%$ compliance gap between the two finest meshes), and the optimality-criteria convergence metric does not decrease monotonically.

Ensemble Kalman Filtering for the Chaotic Lorenz-96 System

[GitHub repository](#)

- Implemented and independently validated a stochastic Ensemble Kalman Filter (perturbed observations) for the chaotic Lorenz-96 system ($K = 40$, $F = 8$) against a linear/Gaussian Kalman-filter problem; estimated the largest Lyapunov exponent $\lambda_1 \approx 1.69$ via Benettin renormalisation.
- Diagnosed severe ensemble under-dispersion and filter collapse at small ensemble sizes, and reduced assimilation RMSE by over $11\times$ using Gaspari–Cohn covariance localisation, quantifying ensemble-size, inflation, and observation-density/noise effects.

Applied ML & Engineering

INTUOS Flight Data Monitoring Dashboard: Full-stack aviation analytics platform (FastAPI + IBM DB2, React/Vite, Docker) serving ML-based flight-phase classifiers (5-fold weighted F1 0.999) over recorded telemetry; deployed at Intuos Srl.

LLM Engineering Portfolio: Nine end-to-end LLM/generative-AI systems, including a RAG assistant, an NL-to-SQL pipeline, and a pruning-and-distillation pipeline for a FinBERT classifier (96.9% accuracy/F1 at 88M params vs. 97.6%/0.976 at 110M).

Full portfolio, including additional applied-math and engineering projects: adebanjioluwatimileyin.github.io/projects.html

TECHNICAL SKILLS

Mathematical / Numerical: Numerical PDEs, finite-element methods, pseudo-spectral methods, variational methods, reduced-order modelling (POD–Galerkin), hyper-reduction (DEIM), inverse problems, PDE-constrained optimisation, Tikhonov/variational regularisation, Bayesian inference (MCMC/pCN), ensemble data assimilation

Scientific ML / ML: PyTorch, TensorFlow, scikit-learn, physics-informed neural networks, Fourier Neural Operators, surrogate/operator learning, supervised learning, model evaluation

Programming / Scientific Computing: Python, NumPy, SciPy, MATLAB, C++, FEniCSx, PETSc, JAX, FFT-based computing

Engineering / Data: SQL, IBM DB2, PostgreSQL, FastAPI, Docker, Git, REST APIs, ETL/data pipelines, AWS

EDUCATION

2018 – 2021

M.Sc. Mathematical Engineering | University of L’Aquila, Italy | Final grade: 106/110

Thesis: “Bounds on mixing of passive scalars advected by incompressible energy and enstrophy constrained flows and Anomalous dissipation” ([code and report](#)) – derived and numerically validated, via a pseudo-spectral solver, mixing-rate bounds for passive scalars in incompressible, enstrophy-constrained flow. *Supervisor:* Prof. Stefano Spirito.

Relevant coursework: Functional Analysis, Applied PDEs, Fluid Dynamics, Numerical Methods for PDEs, Finite Element Methods, Scientific Computing, Dynamical Systems and Bifurcation Theory, Advanced Optimisation.

2014 – 2018

B.Sc. Mathematics | Obafemi Awolowo University, Nigeria | First Class Honours

Thesis: Application of Galerkin Finite Element Method to the One-Dimensional Stefan Problem.

RESEARCH EXPERIENCE

2020 – 2021

M.Sc. Thesis Research | University of L'Aquila

Investigated mathematical bounds for passive-scalar mixing in incompressible, enstrophy-constrained flows. Derived analytical bounds and implemented a pseudo-spectral Python solver to validate theoretical predictions computationally.

2017 – 2018

B.Sc. Thesis Research | Obafemi Awolowo University

Applied the Galerkin finite-element method to the one-dimensional Stefan free-boundary problem, developing numerical algorithms for phase-boundary tracking.

AWARDS & LEADERSHIP

Awards

University of L'Aquila International Scholarship

2018 – 2021

Professor S.S. Okoya Prize for Excellence in Applied Mathematics

2018

MTN Foundation Scholarship

2014 – 2018

PTDF Scholarship

2014 – 2018

Murli T. Chellaram Foundation Scholarship

2014 – 2018

Teaching & Leadership

2016 – 2018

Founder & President, Inter-University Mathematics Club | Nigeria

Established a cross-university mathematics mentorship network and organised seminars on mathematics, research, and academic development.

2015 – 2018

Academic Director, NAMSS | **Peer Tutor** | Obafemi Awolowo University

Coordinated workshops and tutored undergraduate students in mathematical analysis and quantitative reasoning.

REFERENCES

Prof. Ogundare Babatunde Sunday
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Dr. Yusuf Tijani
Lecturer, University of the Witwatersrand, South Africa
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